

WPR — Adaptive Ribbon Source

Version 1.0.1 · Audio Unit and VST3

What WPR does

WPR is a pressure-responsive ribbon source stage for a user-provided microphone. It models proximity, ribbon movement, polar behavior, transformer/iron color, and protected output. It is designed to be useful with inexpensive and high-end microphones alike, but it does not identify a microphone model or price from audio and it does not apply an undisclosed “flat” correction. Exact frequency correction requires a measured calibration profile or reference recording.

Signal flow is: input and channel mode → proximity → mechanical pressure/excursion → ribbon element → polar/stereo field → transformer and iron core → protected output.

First-use workflow

1. Insert WPR on the microphone track and use the host meter to set healthy headroom. 2. Choose Mono, Stereo, or Dual routing to match the source and session. 3. Set the pickup behavior with Proximity, Width, Focus, and Polar Pattern. Check mono compatibility before widening. 4. Shape the ribbon movement with the mechanical controls below. 5. Add transformer/iron color only after the ribbon response is stable, then level-match the result against bypass.

Mechanical pressure controls

These controls respond to the incoming pressure envelope. They are not a static low-cut and they are not a conventional compressor.

Control	Range	Default	Function
Ribbon Damping (DAMPING)	0–1	0.618034	Recovery of the pressure state. Higher values settle faster; lower values retain movement.
Ribbon Excursion (EXCURSION)	0–1	0.236068	Amount of pressure-driven ribbon travel and transient movement.
Ribbon Guard (RIBBON_GUARD)	Off / Soft / Medium / Hard	Off	Progressive protection against rise-driven excursion. Use Soft or Medium for plosives and aggressive transients; reserve Hard for protection, not tone.

The remaining controls cover ribbon tone, mass, blend, air edge, proximity, stereo width/focus, polar pattern, transformer drive, magnetic flux, headroom, hysteresis, winding/core choices, and output trim. Every control is automatable and recalled with the DAW session.

Microphone identity and calibration

WPR is the adaptive ribbon stage in the DIHWorld vocal/mic family. SILVER is the neutral M50-referenced comparison stage; GOLD uses an M50-referenced input followed by deliberate harmonic color. These are role labels, not price tiers. A signal-only plugin cannot know whether a microphone is \$50 or \$5,000. If a production needs exact corrective EQ for a physical microphone, measure that microphone and build a profile; WPR can then provide the pressure/ribbon behavior around that calibrated source.

Metering and routing

The editor shows a relative peak meter and stereo-field readout. They are observation aids; the DAW's input, output, and loudness meters remain authoritative. WPR accepts matched mono/stereo layouts and mono-in to stereo-out. In wider matched layouts, only the first two channels are processed and extra channels pass through.

Troubleshooting

- **No obvious change:** confirm WPR is enabled, raise Ribbon Excursion slightly, and compare at matched output level.
- **Plosives or bursts:** lower Excursion or engage Soft/Medium Ribbon Guard; do not compensate by driving the transformer harder.
- **Too slow or dark:** raise Damping and reduce Ribbon Mass or Proximity before adding high-frequency color.
- **Stereo feels unstable:** use a narrower Polar Pattern or Focus, then verify the signal in mono.
- **Need exact mic correction:** provide a measured calibration/reference profile; WPR cannot infer model, price, or hidden flat response from the waveform alone.

Quit the DAW before replacing an installed AU/VST3. If the new build is not listed, clear the Audio Unit cache and relaunch the host.